ESP32-WROOM-32E BLE LED Control Guide

Control an LED from an iPhone using nRF Connect, LightBlue, or Bluefruit Connect.

Hardware Wiring

GPIO 2 -> 220-330 Ohm Resistor -> LED Anode (+); LED Cathode (-) -> GND

Arduino Sketch

```
#include
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#define LED_PIN 2
#define SERVICE_UUID "12345678-1234-1234-1234-123456789abc"
#define CHARACTERISTIC_UUID "abcdef01-1234-1234-1234-123456789abc"

BLECharacteristic *pCharacteristic;

class MyCallbacks : public BLECharacteristicCallbacks {
void onWrite(BLECharacteristic *pCharacteristic) {
String value = pCharacteristic->getValue().c_str();
if (value == "1") digitalWrite(LED_PIN, HIGH);
else if (value == "0") digitalWrite(LED_PIN, LOW);
}
};

void setup() {
pinMode(LED_PIN, OUTPUT);
BLEDevice::init("ESP32 LED Control");
BLEServer *pServer = BLEDevice::createServer();
BLEService *pService = pServer->createService(SERVICE_UUID);
pCharacteristic = pService->createCharacteristic(CHARACTERISTIC_UUID,
BLECharacteristic::PROPERTY_READ | BLECharacteristic::PROPERTY_WRITE);
pCharacteristic->setCallbacks(new MyCallbacks());
pService->start();
BLEDevice::getAdvertising()->start();
}

void loop() {}
```

Using nRF Connect or LightBlue

• Scan for device: ESP32 LED Control

• Connect to the device.

• Open service UUID 12345678-1234-1234-1234-123456789abc.

• Open characteristic UUID abcdef01-1234-1234-1234-123456789abc.

• Write 1 to turn the LED ON.

• Write 0 to turn the LED OFF.

Optional Enhancement

Modify the code to accept values 0-255 for PWM brightness control.